

Work Orders Dataset

Data Specification

June 2026

Overview

This dataset contains **44,265 work orders** from facility management operations at two Valmet industrial sites in Finland. The data spans **May 2017 – June 2026** (~9 years) and covers cleaning services, property maintenance, technical services, and facility services.

The dataset is **anonymized**: all personal names, phone numbers, and email addresses in free-text fields have been replaced with [NAME], [PHONE], and [EMAIL] tokens. All other content is preserved.

Language: Structured columns are bilingual (Finnish + English). Free-text fields are in **Finnish**.

File Details

Property	Value
Filename	work_orders_anonymized.csv
Format	CSV (UTF-8 with BOM)
Rows	44,265 (excluding header)
Columns	32
Date range	2017-05-29 – 2026-06-01
Delimiter / Text qualifier	Comma (,) / Double quote (")

Sites

Customer	Site	Address	Count
Valmet Flow Control Oy	Venttiilitehdas	Vanha Porvoontie 229, Vantaa	25,744
Valmet Technologies Oy	Lentokentänpolku 11	Lentokentänpolku 11, Tampere	18,330
Valmet Flow Control Oy	Toimistotalo	Vanha Porvoontie 229, Vantaa	151
Valmet Flow Control Oy	STD tehdas	Vanha Porvoontie 229, Vantaa	40

Column Reference

Identifiers & Location

Column	Type	Null?	Description
CUSTOMER_NO	Integer	No	Customer number / ID in Luotea's ERP (2 unique)

Column	Type	Null?	Description
customer_name	Text	No	Customer company name
CUSTOMER_SITE_NO	Integer	No	Site identifier (4 unique)
customer_site_name	Text	No	Human-readable site name
site_address	Text	No	Physical address
CUSTOMER_WORKSITE_NO	Text	No	Worksite sub-identifier (18 unique)

Contract & Service Line

Column	Type	Null?	Description
CONTRACT_NO	Integer	Yes	Contract number
CONTRACT	Text	No	Contract type code (see below)
SERVICE_LINE_FIN	Text	No	Service line in Finnish
SERVICE_LINE_ENG	Text	No	Service line in English

Code	Finnish	English	Count	Share
SP	Siivouspalvelut	Cleaning services	26,616	60.1%
KIPA	Kiinteistöpalvelut	Facility services	7,623	17.2%
KH	Kiinteistöhuolto	Property maintenance	6,473	14.6%
KT	Kiinteistötekniikka	Technical services	3,553	8.0%

Work Order Core Fields

Column	Type	Null?	Description
WO_NO	Integer	No	Primary key. Unique ID for an individual work order.
WORK_ORDER_TYPE	Categorical	No	EH-työ (Scheduled) or Tilaustyö (On-demand)
WORK_ORDER_TYPE_ENG	Categorical	No	English: Scheduled work or On-demand work

Scheduled work (EH- työ, 3,241 rows): Generated from the ERP system based on manual programming. Frequencies and scope are based on the contract and service description.

On-demand work (Tilaustyö, 41,024 rows): Generated by maintenance personnel, supervisors, customer service, through customer systems via integration, or through forms on the Luotea website. May or may not be included in the contract.

Assignment & Priority

Column	Type	Null?	Description
ASSIGNMENT_TYPE	Categorical	Yes	Manuaalinen osoitus or DS-osoitus. N/A for SP.
ASSIGNMENT_TYPE_ENG	Categorical	Yes	Manually assigned or Dynamically scheduled
PRIORITY_ID	Integer	Yes	Priority 1–6 (1=most urgent). N/A for SP.

Manually assigned (5,886): Work order assigned to a maintenance person directly by a supervisor.

Dynamically scheduled (5,935): ERP dynamic scheduling assigned the task automatically based on priority and SLA.

Priority	Count	Meaning
1	365	Most urgent
2	966	Urgent
3	5,189	Normal
4	5,001	Low
5	300	Planned
6	720	Least urgent

Free-Text Fields (Anonymized)

Column	Null?	Avg len	Description
WORK_DESCRIPTION	Yes	114	Longer description. Additional detail beyond the short description.
WORK_ORDER_DESCRIPTION	No	16	Short title (max 60 chars, may be truncated).
WORK_ORDER_PERFORMED_ACTION	Yes	121	Maintenance person's completion comments. Primarily used for KT and KH. Cleaning (SP) does not require this.

Common patterns in WORK_ORDER_PERFORMED_ACTION:

Työn Tilaaaja: [NAME] — orderer | Tilattu työ: ... — ordered work | Tehty työ: ... — completed work | Ilmoittaja: [NAME] [PHONE] — reporter

Work Type Classification

Column	Type	Null?	Description
WORK_TYPE_DESCRIPTION_FIN	Text	No	Detailed work category in Finnish (39 unique)
WORK_TYPE_DESCRIPTION_ENG	Text	No	Detailed work category in English (39 unique)

Scheduled Maintenance (PM)

Column	Type	Null?	Description
PM_NO	Categorical	Yes	PM plan number. Every EH-työ has a PM_NO. Can be used to link to the Scheduled Maintenance dataset. 94 unique.
PM_ACTION_DESCR	Text	Yes	Description of scheduled maintenance activity (33 types)

Timestamps & Duration

Column	Type	Null?	Description
WORK_STARTED_DATETIME	Datetime	No	Format: YYYY-MM-DD HH:MM:SS.000
WORK_FINISHED_DATETIME	Datetime	No	Work finish timestamp
WORK_FINISHED_DAYS	Decimal	No	Duration start-to-finish in days (0–1,261)
WORKTIME_HOURS	Decimal	Yes	Actual hours to complete the task. NULL = subcontractor work or non-standard billing.

SLA

Column	Type	Null?	Description
SLA_REQUIRED_START_DATETIME	Datetime	Yes	Required SLA start date. Applies to EH-työ (Scheduled work).
SLA_REQUIRED_END_DATETIME	Datetime	Yes	Required SLA end date. Applies to EH-työ (Scheduled work).
IS_SLA_VIOLATION	Boolean	No	0/1 flag. Overall violation rate: 53.8%

Billing & Subcontracting

Column	Type	Null?	Description
WORK_INVOICABLE	Boolean	No	True = billable (on-demand, not in contract). False = included in contract, not separately billable. 8.8% billable.
IS_SUBCONTRACTOR_WORK	Boolean	No	True = subcontractor, False = Luotea. 1.9% subcontractor.

Anonymization Tokens

Token	Meaning	Rows affected
[NAME]	Personal name (first + last)	3,698
[PHONE]	Phone number (Finnish format)	865
[EMAIL]	Email address	222

Anonymization applied only to WORK_DESCRIPTION, WORK_ORDER_DESCRIPTION, and WORK_ORDER_PERFORMED_ACTION. All other columns are unmodified.

Work Type Categories (Top 15)

Work Type	Count
Cleaning services	25,248
Technical maintenance and operational maintenance	7,330
Security services	2,142
Workplace and premises services	1,869
Repairs and maintenance electrical	1,701
Outdoor area and green area maintenance	1,688
Additional cleaning services	1,611
Repairs and maintenance heating and water systems	689
Repairs and maintenance ventilation	257
Additional property maintenance services	232
Repairs and maintenance building automation	200
Repairs and maintenance property operation and maintenance	158
Periodic maintenance electrical	125
Winter maintenance of outdoor areas	115
Repairs and maintenance fire safety	100

Notes for Participants

1. **NULL values** appear as the literal string NULL (not empty strings).
2. **Free-text is in Finnish.** Use translation or Finnish NLP if needed.
3. **WORK_ORDER_DESCRIPTION** is max 60 chars and often truncated. Combine with WORK_DESCRIPTION and WORK_ORDER_PERFORMED_ACTION for full context.
4. **SLA violation rate is 53.8%** — may be a useful analysis target.
5. **WORK_FINISHED_DAYS** can reach 1,261 days for long-running or re-opened orders.
6. **Encoding:** UTF-8 with BOM. If the first column name appears garbled, strip the BOM.
7. **ASSIGNMENT_TYPE** and **PRIORITY_ID** do not apply to cleaning services (SP).
8. **PM_NO** can be used to join with the Scheduled Maintenance dataset.